

Interactive HTML Generator

This script generates an interactive HTML visualization with a sunburst chart from the hierarchical question data.

Prerequisites

Python Environment Setup

This project requires Python 3.6 or higher. Follow these steps to set up your Python environment:

1. Check Python Installation

```
python3 --version
```

If Python is not installed, download it from python.org or use your system's package manager:

- **macOS:** `brew install python3`
- **Ubuntu/Debian:** `sudo apt-get install python3 python3-pip python3-venv`
- **Windows:** Download from python.org or use `winget install Python.Python.3`

2. Create a Virtual Environment (Recommended) Using a virtual environment keeps dependencies isolated from your system Python:

```
# Navigate to the project root  
cd /path/to/industria-4.0
```

```
# Create virtual environment  
python3 -m venv venv
```

```
# Activate the virtual environment  
# On macOS/Linux:  
source venv/bin/activate
```

```
# On Windows:  
venv\Scripts\activate
```

You'll know the virtual environment is active when you see (venv) in your terminal prompt.

3. Install Dependencies Install all required Python packages using pip:

```
# Make sure you're in the question-base directory or adjust the path  
pip install -r question-base/requirements.txt
```

Core dependencies installed:

- `python-docx` - Read and write Microsoft Word .docx files
- `jonschema` - JSON Schema validation

- openpyxl - Read/write Excel files
- pyyaml - YAML support
- pandas - Data analysis
- tabulate - Pretty-print tabular data
- colorama - Colored terminal output

4. Verify Installation

Test that imports work

```
python3 -c "import json, sys, pathlib; print('Python environment ready!')"
```

5. Deactivating the Virtual Environment When you're done working, deactivate the virtual environment:

```
deactivate
```

Quick Start (After Setup)

Activate virtual environment

```
source venv/bin/activate
```

Navigate to scripts directory

```
cd question-base/scripts
```

Run the HTML generator

```
python generate_index_html.py ../JSON7_20260105_164046/metadata/hierarchy_table.md
```

Overview

generate_index_html.py reads the markdown hierarchy table and creates a beautiful, interactive single-page application that allows you to explore all Industry 4.0 maturity model questions through a sunburst chart.

Features

- **Interactive Sunburst Chart:** Visual representation of the hierarchical structure (Block → Pilar → Dimension → Capacity → Question)
- **Click to Explore:** Click on any segment to view question details
- **Beautiful Design:** Modern, responsive UI with gradient backgrounds
- **Question Details:** Full question information displayed in the bottom half
- **Random Start:** Shows a random question on page load to get you started
- **Breadcrumb Navigation:** Shows the full hierarchy path for each question

Usage

First: Rebuild Hierarchy

After validation moves files or JSON files get modified by hand, rebuild the hierarchy to reflect the new structure.

```
OUTPUT_DIR="../../JSON7_20260105_164046"  
# Rebuild hierarchy.json  
python rebuild_hierarchy.py "$OUTPUT_DIR"
```

Output:

– Updated \$OUTPUT_DIR/metadata/hierarchy.json with correct statistics

Expected statistics (as of 2025-12-26):

□ Hierarchy table saved to: ../../JSON7_20260105_164046/metadata/hierarchy_table.md

Statistics:

Total capacities: 23
Total questions: 137
Total dimensions: 15
Total pilares: 7
Total blocks: 3

Basic Usage

```
python generate_index_html.py <path_to_hierarchy_table.md>
```

The HTML file will be created as index.html in the same directory as the markdown file.

Specify Output Path

```
python generate_index_html.py hierarchy_table.md -o custom_output.html
```

Complete Workflow Example

1. Convert all JSON files into an intermediary hierarchy table

```
OUTPUT_DIR="../../JSON7_20260105_164046"  
python rebuild_hierarchy.py "$OUTPUT_DIR"
```

2. Generate interactive HTML

```
python generate_index_html.py $OUTPUT_DIR/metadata/hierarchy_table.md
```

3. Open in browser

```
open $OUTPUT_DIR/metadata/index.html
```

Input Format

The script expects a markdown table with the following columns:

Block	Pilar	Dimension	Capacity	Question Code	Question Title
...	...	...	...	...	...

This is automatically generated by `batch_convert.py` when you run the batch conversion.

Output

The script generates a single HTML file (`index.html`) that:

- Contains all the question data embedded as JSON
- Uses `D3.js` (loaded from CDN) for the sunburst visualization
- Is completely self-contained and can be opened directly in any modern browser
- Requires no web server or additional dependencies

Technical Details

Dependencies

- **Python 3.6+:** For running the script
- **D3.js v7:** Loaded from CDN in the generated HTML (no local install needed)

Browser Support

The generated HTML works in all modern browsers:

- Chrome/Edge (recommended)
- Firefox
- Safari

File Size

The generated HTML file size depends on the number of questions:

- ~58KB for 140 questions
- Scales linearly with question count

Color Scheme

The sunburst chart uses a color gradient to distinguish hierarchy levels:

- **Block:** Purple (`#667eea`)
- **Pilar:** Deep Purple (`#764ba2`)

- **Dimension:** Pink Gradient (#f093fb)
- **Capacity:** Blue Gradient (#4facfe)
- **Question:** Green Gradient (#43e97b)

Example Output

The generated index.html provides:

Top Half: Interactive Sunburst Chart

- Click on any segment to drill down
- Hover to see tooltip with full hierarchy path
- Responsive sizing that adapts to window size

Bottom Half: Question Details

- Breadcrumb showing: Block → Pilar → Dimension → Capacity
- Question ID badge
- Question title (large, bold)
- Full hierarchy context

Troubleshooting

No questions found

- Make sure the markdown file has the correct table format
- Check that the table has data rows (not just headers)

HTML file not opening

- Try opening with file:// protocol or through a web server
- Some browsers may block local file access to CDN resources

Chart not displaying

- Check browser console for errors
- Ensure internet connection (needed for D3.js CDN)
- Try a different browser

Future Enhancements

Potential improvements:

- Add search/filter functionality
- Export selected questions
- Print-friendly view
- Dark mode toggle
- Zoom/pan controls for chart
- Integration with maturity level data from JSON files

License

Part of the Industry 4.0 Maturity Model project.